

MA 102 Mathematics II
Ordinary Differential Equations
Lecture 6
(March 13, 2024)



The solution of a first-order ordinary differential equation contains an arbitrary constant. Consider the following first-order ODE:

$$\frac{dy}{dx} = f(x, y). \quad (1)$$

The solution of (1) can be written in functional form as

$$F(x, y, c) = 0. \quad (2)$$

Equation (2) represents a curve in the xy -plane for a specific value of the constant c . For different values of c , we can generate as many curves as we wish. In totality, we have a family of curves.

In many applications of interest, it is of considerable significance to determine a family of curves which is mutually perpendicular to a given family of curves. Then, the curves of these two families are said to be **mutually orthogonal** and the curves of the family to be determined are called **orthogonal trajectories** of the given curves.

Mathematically, the orthogonal trajectories of the solution of a first-order ODE $y' = f(x, y)$ are described by the solution of the ODE $y' = -1/f(x, y)$ implying that both the solutions (or both the families) intersect at right angles.

Find the orthogonal trajectories of the family of circles $x^2 + y^2 = a^2$.

The given family is

$$x^2 + y^2 = a^2. \quad (3)$$

Differentiating (3) with respect to x , we get the differential equation as

$$\begin{aligned} x + yy' &= 0 \\ \Rightarrow y' &= -x/y. \end{aligned}$$

The slope of the family orthogonal to the given family (3) will be given by $y' = y/x$. Solving it, we will get $y = cx$ which is a family of straight lines passing through the origin with different constant slopes and perpendicular to the given family of circles.



The path of a particle of fluid is called a **streamline** and the orthogonal trajectories are called **equipotential lines**. If the streamlines are given by $xy = c$, find the equipotential lines.

The given family is

$$xy = c. \quad (4)$$

Differentiating (4) with respect to x , we get the differential equation as

$$\begin{aligned} xy' + y &= 0 \\ \Rightarrow y' &= -y/x. \end{aligned}$$

The slope of the family orthogonal to the given family (4) will be given by $y' = x/y$. Solving it, we will get $x^2 - y^2 = k^2$ which is a family of hyperbolas representing the equipotential lines and perpendicular to the given family of rectangular hyperbolas (streamlines).

Find the orthogonal trajectories of the family of parabolas $y^2 = cx$.

The given family is

$$y^2 = cx. \quad (5)$$

Differentiating (5) with respect to x , we get

$$2yy' = c$$

Putting $c = y^2/x$ above, we get the differential equation for the family as

$$y' = y/(2x).$$

The slope of the family orthogonal to the given family (5) will be given by $y' = -(2x)/y$. Solving it, we will get $y^2 + 2x^2 = k$ which is a family of ellipses and perpendicular to the given family of parabolas.



The first-order equations can be applied in solving problems arising in engineering and applied sciences such in mechanics, electrical circuits, chemistry, biology, economics, etc. We discuss some of them below.

Recall Newton's law of cooling:

The rate at which the temperature of an object changes is proportional to the difference between the instantaneous temperature of the object and the temperature in the surrounding.

Mathematically, we have

$$\frac{dT}{dt} = -k(T - S), \quad k > 0, \quad T(t) > S. \quad (6)$$

Example

If an object whose temperature is initially 100°C is allowed to cool in air which remains at the constant temperature 20°C , and if it is observed that, in 10 minutes, the object has cooled to 60°C , find the instantaneous temperature of the object.

With $S = 20^{\circ}\text{C}$, we have the governing differential equation as

$$\frac{dT}{dt} = -k(T - 20), \quad k > 0, \quad T(t) > 20. \quad (7)$$

Separating the variables T and t , the general solution of (7) can be obtained as

$$T(t) = ce^{-kt} + 20. \quad (8)$$

Imposing the initial condition $T(0) = 100$,

we can find c as $c = 80$. In turn, the solution can now be written as

$$T(t) = 80e^{-kt} + 20. \quad (9)$$

To obtain the proportionality constant k ,

we use the other available condition $T(10) = 60$ to get $60 = 80e^{-10k} + 20$ giving $10k = \ln 2$, i.e., $k = 0.0693$. Hence, the solution to the given problem can be written as

$$T(t) = 80e^{-0.0693t} + 20.$$

Problems of growth and decay:

The differential equation

$$\frac{dy}{dt} = ky \quad (10)$$

may look very simple but its solutions gives us information about growth and decay of an object/species. The law states that the rate of change of a quantity y is proportional to the quantity itself.

If $k > 0$ and $y > 0$, then it is obvious that $\frac{dy}{dt} > 0$ and we say that the solution y is increasing. It makes the problem a **problem of growth**.

On the other hand, if $k < 0$ and $y > 0$, then $\frac{dy}{dt} < 0$ and we say that the solution is decreasing. It makes the problem a **problem of decay**.

This law applies to many real-life scenarios, for example, the growth of bacteria and the decay of a radioactive substance (as time progresses). The general solution is usually very easy to find for such simple equations. Finding the particular required solution depends on the given condition(s).



Example:

In a culture of bacteria, the rate of growth is proportional to its population $y(t)$ present at time t with the population doubling in one day. How much bacteria can be expected after seven days at the same rate of growth?

Solution:

$y(t)$ is the number of bacteria present at time t . We have the governing equation for its growth as

$$\frac{dy}{dt} = ky, \quad k > 0. \quad (11)$$

Its solution can be found easily as

$$y(t) = ce^{kt}. \quad (12)$$

We assume that the initial population of bacteria is y_0 , i.e., the initial condition $y(0) = y_0$.

Using this condition in (12), we get the solution as

$$y(t) = y_0 e^{kt}, \quad (13)$$

According to the given condition, we also have $y(1) = y_0 e^k = 2y_0$

This gives $k = \ln 2 = .693$. The solution to problem (11) can be written as

$$y(t) = y_0 e^{(\ln 2)t}. \quad (14)$$

To address the last part,

$$\begin{aligned} y(7) &= y_0 e^{(\ln 2)(7)} \\ &= y_0 e^{\ln 2^7} \\ &= 2^7 y_0. \end{aligned}$$

This means that, after 7 days, the population of bacteria will be 2^7 times its initial population.

In a similar manner, the decay problems can be discussed. In this case, the governing equation will be

$$\frac{dy}{dt} = -ky, \quad k > 0. \quad (15)$$

Its solution is

$$y(t) = ce^{-kt}. \quad (16)$$

Example:

Experiments show that most radioactive substances disintegrate at a rate proportional to the amount of the substance present at that instant. The time in which 50% of the given amount disappears is called the half-life of the substance. It is found that the half-life for radium is 1600 years. What percent of radium will disappear in 1 year and in 100 years?

Solution

$$y(t) = y_0 e^{-0.000433t},$$

with y_0 being the initial amount of radium present.

0.1% disappears in one year.

4.3% disappears in 100 years.

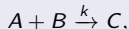
Chemical reactions

A first-order chemical reaction is defined when, in a chemical reaction, the rate of change of concentration of the chemical species is proportional to the concentration. For the typical first-order irreversible decomposition, we can write the chemical reaction as $A \xrightarrow{k} \text{products}$.

If a is the concentration of the species A at time $t = 0$, and x is the concentration at time t , then a first-order chemical reaction is represented by the following first-order differential equation:

$$\frac{dx}{dt} = k(a - x). \quad (17)$$

In a similar manner, we can define the second-order reaction or bimolecular reaction as



where molecules of species A and B react to form molecules of species C .

If the original concentrations of A and B are given by a and b , respectively, and if x represents the concentration of C at time t , then the second-order reaction can be represented by the following first-order ODE:

$$\frac{dx}{dt} = k(a - x)(b - x). \quad (18)$$

To solve (17):

Separating the variables and integrating, we get

$$a - x = de^{-kt}. \quad (19)$$

Using the fact that $x(0) = 0$, we get $d = a$.

Hence, the solution of (17) can be obtained as

$$x(t) = a(1 - e^{-kt}). \quad (20)$$

To solve (18):

Separating the variables and adjusting, we get

$$\frac{1}{b-a} \left(\frac{1}{a-x} - \frac{1}{b-x} \right) dx = kdt.$$

Upon solving:

$$\frac{b-x}{a-x} = de^{(b-a)kt}.$$

Using the fact that $x(0) = 0$, we get $d = b/a$.

Hence, the solution of (18) can be obtained as

$$\frac{b-x}{a-x} = \frac{b}{a} e^{(b-a)kt},$$

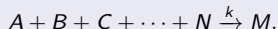
which, on simplification, becomes

$$x(t) = \frac{ab[1 - e^{(b-a)kt}]}{a - be^{(b-a)kt}}.$$

This is the concentration of the product C at any time t .



In a similar manner, we can extend this notion to a reaction of any order. Consider the following reaction of n components:



The associated differential equation is given by

$$\frac{dx}{dt} = k(a - x)(b - x)(c - x) \cdots (n - x). \quad (21)$$

Next, we look into mathematical models governing electric circuits.

The simplest electric circuit is a series circuit in which we have a source of electric energy (electromotive force, i.e., emf) such as a generator or a battery, and a resistor, which uses energy such as an electric bulb, fan or any type of appliance. The notations are

I : current

R : resistance

L : inductance

E : emf.



Kirchoff's Voltage Law:

The algebraic sum of all the instantaneous voltage drops around any closed loop is zero, or the voltage supplied is equal to the sum of the voltage drops.

Note the following for an RL circuit:

- voltage drop across resistance $= RI$
- voltage drop across inductance $= L \frac{dI}{dt}$
- voltage supplied $= E(t)$.

Using Kirchoff's law, we have the following first-order equation in I :

$$L \frac{dI}{dt} + RI = E(t). \quad (22)$$

Now two cases arise:

- **Case I:** Voltage is constant, i.e., $E(t) = E_0$
- **Case II:** Voltage is periodic, say, $E(t) = E_0 \cos \omega t$.

Case I: Voltage is constant, i.e., $E(t) = E_0$

Equation (22) takes the form

$$L \frac{dI}{dt} + RI = E_0. \quad (23)$$

Considering the integrating factor $e^{Rt/L}$, equation (23) gives the solution

$$I(t) = Ae^{-Rt/L} + \frac{E_0}{R}. \quad (24)$$

If there is no current at $t = 0$, we have the initial condition $I(0) = 0$. It gives us $A = -E_0/R$.

Hence, the solution becomes

$$I(t) = \frac{E_0}{R} (1 - e^{-Rt/L}). \quad (25)$$

It can be clearly observed from (25) that, as $t \rightarrow \infty$, the current $I(t) \rightarrow \frac{E_0}{R}$ which represents a **steady-state solution**.

Case II: Voltage is periodic, say, $E(t) = E_0 \cos \omega t$.

Equation (22) takes the form

$$L \frac{dI}{dt} + RI = E_0 \cos \omega t. \quad (26)$$

Considering the integrating factor $e^{Rt/L}$, equation (26) gives the solution

$$I(t) = e^{-Rt/L} \left\{ A + \frac{E_0}{L} \int e^{Rt/L} \cos \omega t \, dt \right\}. \quad (27)$$

Recall the integration formula

$$\int e^{at} \cos bt \, dt = \frac{e^{at}}{a^2 + b^2} (a \cos bt + b \sin bt).$$

Applying this formula, we get the solution as

$$I(t) = Ae^{-Rt/L} + \frac{E_0}{R^2 + \omega^2 L^2} (R \cos \omega t + L\omega \sin \omega t). \quad (28)$$

Here, the first term is called the transient solution (it dies out as $t \rightarrow \infty$). The second part is called the steady-state solution because, after a long time, the current $I(t)$ executes periodically harmonic oscillations.

Using the initial condition $I(0) = 0$, we can evaluate A as

$$A = -\frac{E_0 R}{R^2 + L^2 \omega^2}.$$

The solution can now be written as

$$I(t) = -\frac{E_0 R}{R^2 + L^2 \omega^2} e^{-Rt/L} + \frac{E_0}{R^2 + L^2 \omega^2} (R \cos \omega t + L \omega \sin \omega t). \quad (29)$$

In a similar manner, the current or charge problem in an RC circuit (C : Capacitance) can be tackled by considering the following ODE:

$$R \frac{dQ}{dt} + \frac{Q}{C} = E(t). \quad (30)$$

Once $Q(t)$ is found, the current $I(t)$ can be found by evaluating $\frac{dQ}{dt}$.